

Chapter 5

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Math in Action: Chapter 5 Worksheet

Hey, math champs! This quick worksheet is your chance to tackle real-world puzzles, crack codes, and design epic stuff using skills from Chapter 5. From budgeting a party to predicting game odds, let's make math your superpower. Complete these mini-missions solo or with a squad—show off your skills in 30 minutes or less!

Mission 1: Budget Blast (Algebra)

Task: Plan a "Movie Night" for 4 friends with a \$50 budget. Tickets are \$8 each, and popcorn is \$3 per person.

- Write an equation for total cost (t) based on people (p).
- Solve: Can you afford it? If not, how many friends can you invite?
- **Answer Space:**

Mission 2: Design Dash (Geometry)

Task: Sketch a mini "Gaming Nook" in a 10 ft x 8 ft room corner. Include a rectangular desk (3 ft x 2 ft).

- Calculate desk area and remaining room area.
- Bonus: Add a circular rug (radius 2 ft). What's its area? (Use $\pi \approx 3.14$)
- **Answer Space & Sketch:**

Mission 3: Data Detective (Data Analysis)

Task: Track a fun stat for 3 days (like "hours gaming" or "snacks eaten"). Example data: Day 1: 2 hrs, Day 2: 3 hrs, Day 3: 1 hr.

- Find the average. Make a tiny bar graph (draw bars for each day).
- What trend do you see?
- **Answer Space & Graph:**

Mission 4: Odds Oracle (Probability)

Task: You're in a raffle with 20 tickets, and 2 win prizes.

- What's your probability of winning if you buy 1 ticket? Express as a fraction and percentage.
- Bonus: If you buy 5 tickets, how do your odds change?
- **Answer Space:**

Mission 5: Code Craze (Coding Logic)

Task: Write pseudocode (plain English steps) for a "Homework Timer" app.

- Example: If time spent $\geq$ 2 hours, print "Keep Going!"; Else, print "Break Time!"
- Add one fun feature (like a reward message).
- **Answer Space:**

Bonus Fun Twist: Math Meme Mania

Task: Create a quick, funny meme or one-liner about math solving a teen problem (like "Algebra: Splitting pizza bills since forever!").

- Write or sketch it here:

Reflection: What's one way you'll use math IRL this week (like budgeting allowance or sizing up a new poster)? Jot down 1-2 sentences.